

Composing music

Portfolio assignment 3

MAL2 Autumn 2025

In this assignment, you will implement, train and evaluate **at least two different character-based language models** (*i.e.*, char-RNNs) to compose music.

You are free to decide what type of music you wish to compose – whether it is [Lo-Fi hip hop](#) for your studying, the new [national anthem](#) for when AI takes over the world, or something else, you can find data for it. MIDI files are the easiest to work with, and you can find plenty of inspiration online for how to work with them.

As stated above, you should construct at least two different models. The models may differ in any way you find relevant, but some suggestions are:

- GRU vs LSTM vs SimpleRNN cells
- sequence-to-vector vs. sequence-to-sequence models
- plain one-hot encoding vs trainable embeddings
- ... and so on

You are to hand in a notebook with

- output (all cells must be run)
- relevant comments describing your approach, experiments, and findings
- your two language models
- at least four audio files (e.g., .mp3 files) containing examples of your composed music for both models at two different temperatures – these **must be** separate audio files, not audio clips embedded in the notebook
- at least two interesting figures or animations with relevance to the assignment